# CYCLESAFE: MICROSCOPIC TRAFFIC SIMULATION AND INSTRUMENTED BICYCLE TELEMETRY PLATFORM FOR URBAN CYCLIST SAFETY ANALYSIS

**AN UNDERGRADUATE PROJECT REPORT SUBMITTED BY**


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**(Admission No. 23JE0484)**


*Under the Supervision of*

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Assistant Professor, Department of Computer Science and Engineering

**Submitted in partial fulfilment of the requirements for the award of**

**BACHELOR OF TECHNOLOGY**

in

**COMPUTER SCIENCE AND ENGINEERING**


**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**

**INDIAN INSTITUTE OF TECHNOLOGY (INDIAN SCHOOL OF MINES), DHANBAD**


**NOVEMBER, 2026**

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**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**
**INDIAN INSTITUTE OF TECHNOLOGY (INDIAN SCHOOL OF MINES) DHANBAD**

# CERTIFICATE

This is to certify that the project report entitled **"CycleSafe: Microscopic Traffic Simulation and Instrumented Bicycle Telemetry Platform for Urban Cyclist Safety Analysis"** is a bona fide record of the work carried out by **Kodati Tej Deep** (Admission No. **23JE0484**) under my supervision in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology** in **Computer Science and Engineering** at the **Indian Institute of Technology (Indian School of Mines), Dhanbad** during the academic year 2026–2027.

The results embodied in this project report have reached the standard required for submission and have not been submitted to any other university or institute for the award of any degree or diploma.

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**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**
**INDIAN INSTITUTE OF TECHNOLOGY (INDIAN SCHOOL OF MINES) DHANBAD**

# CANDIDATE'S DECLARATION

I hereby declare that the work presented in this undergraduate project report entitled **"CycleSafe: Microscopic Traffic Simulation and Instrumented Bicycle Telemetry Platform for Urban Cyclist Safety Analysis"** is an authentic record of my own research and engineering implementation carried out under the guidance of **Prof. Debasree Das**, Assistant Professor, Department of Computer Science and Engineering, Indian Institute of Technology (Indian School of Mines), Dhanbad.

All sources of empirical data, open-source libraries, academic publications, simulation models, and hardware references have been duly cited and acknowledged in the bibliography. This work has not been submitted in part or full to any other university or institute for the award of any degree, diploma, or qualification.

Date: _____

Place: IIT (ISM) Dhanbad

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**Kodati Tej Deep**

Admission No. **23JE0484**

B.Tech, Computer Science and Engineering

# ABSTRACT

Vulnerable road users, particularly urban cyclists, face disproportionate traffic safety risks from unsafe motor vehicle overtaking maneuvers and aggressive close-passing interactions. Existing safety evaluations predominantly depend either on retrospective police crash records (which suffer from severe under-reporting of near-misses) or on crowd-sourced binary incident markers that lack objective physical distance measurements and vehicle-specific conflict severity weighting.

This project presents **CycleSafe**, an integrated hardware telemetry and microscopic traffic simulation framework for multi-modal urban cyclist safety analysis. The physical telemetry subsystem builds upon the OpenBikeSensor (OBS) architecture, employing an ESP32 microcontroller, dual waterproof ultrasonic distance transducers, u-blox NEO-6M satellite positioning, and an MPU-6050 six-axis inertial measurement unit. We formulate an analytical roll-angle tilt compensation model ($\phi = \arctan(a_y/\sqrt{a_x^2 + a_z^2})$) that eliminates cornering-induced distance measurement distortions, accompanied by a dedicated 3.3 V power regulation rail that prevents SD card data corruption.

In the simulation domain, we develop an empirical microscopic digital twin using Eclipse SUMO and TraCI over a 9.53 km urban corridor in Berlin (Ernst-Reuter-Platz to Oberbaumbrücke). The traffic population is calibrated against empirical 12-hour census data (13,531 motor vehicles and 1,397 bicycles), modeling mixed-lane interactions across passenger cars, heavy-duty trucks, motorcycles, and peer cyclists. We train an extreme gradient boosting (XGBoost) classifier on the TU Berlin SimRA dataset to evaluate conflict encounters across three distinct severity classes (*Safe Pass*, *Close Pass (Medium)*, and *Severe Incident*), incorporating German StVO § 5(4) 1.5 m statutory clearance thresholds and counterpart vehicle mass penalties.

The platform features a full-stack dashboard built with Go, PostgreSQL, Redis, Leaflet, and Chart.js, providing interactive spatial clustering, multi-bicycle telemetry tracking, and one-click CSV export. CycleSafe provides municipal traffic planners and researchers with an objective, reproducible pipeline bridging physical IoT sensing with synthetic urban scale simulation.

**Keywords:** Vulnerable Road Users (VRUs), Bicycle Safety, Microscopic Traffic Simulation, SUMO, OpenBikeSensor, Close-Pass Detection, XGBoost, Inertial Roll Compensation.

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I am sincerely thankful to the **Head of the Department** and all faculty members of the Department of Computer Science and Engineering for providing the academic infrastructure, computing facilities, and constructive learning environment that made this work possible.

I also acknowledge the global open-source communities behind **Eclipse SUMO (Simulation of Urban MObility)**, the **OpenBikeSensor (OBS)** hardware initiative, and the **SimRA** research team at Technische Universität Berlin for making their simulation suites, CAD blueprints, and empirical sensor datasets openly accessible to researchers worldwide.

Finally, I convey my heartfelt thanks to my family and friends for their enduring patience, encouragement, and moral support throughout my academic journey.

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# ABBREVIATIONS AND SYMBOLS

## List of Abbreviations

| Abbreviation | Full Form / Definition |
|---|---|
| **BTP** | Bachelor of Technology Project |
| **CSE** | Computer Science and Engineering |
| **ESP32** | Espressif Systems 32-bit Dual-Core Microcontroller |
| **FDM** | Fused Deposition Modeling (3D Printing) |
| **GPIO** | General Purpose Input/Output |
| **GPS** | Global Positioning System |
| **I2C** | Inter-Integrated Circuit Serial Bus |
| **IDM** | Intelligent Driver Model (Car-Following Physics) |
| **IMU** | Inertial Measurement Unit (MPU-6050) |
| **IoT** | Internet of Things |
| **NMEA** | National Marine Electronics Association 0183 Protocol |
| **OLED** | Organic Light Emitting Diode |
| **OBS** | OpenBikeSensor Open Hardware Standard |
| **OSM** | OpenStreetMap Geographic Database |
| **PET** | Post-Encroachment Time |
| **PETG** | Polyethylene Terephthalate Glycol Filament |
| **PLA** | Polylactic Acid Filament |
| **SPI** | Serial Peripheral Interface |
| **StVO** | Straßenverkehrs-Ordnung (German Road Traffic Act) |
| **SUMO** | Simulation of Urban MObility |
| **ToF** | Time-of-Flight (Acoustic Distance Measurement) |
| **TraCI** | Traffic Control Interface (SUMO Python API) |
| **TTC** | Time-to-Collision |
| **VRU** | Vulnerable Road User |
| **XGBoost** | Extreme Gradient Boosting Machine Learning |

# List of Mathematical Symbols

| Symbol | Meaning / Definition [Units] |
|---|---|
| $d_{\text{measured}}$ | Raw acoustic distance recorded by ultrasonic transducer [m] |
| $d_{\text{true}}$ | Perpendicular ground clearance corrected for frame roll tilt [m] |
| $\phi$ | Instantaneous bicycle frame roll / tilt angle [rad or deg] |
| $a_x, a_y, a_z$ | Tri-axial accelerometer readings from MPU-6050 [m/s$^2$] |
| $v_{\text{ego}}$ | Longitudinal speed of the ego bicycle [m/s or km/h] |
| $v_{\text{other}}$ | Instantaneous speed of overtaking motor vehicle [m/s or km/h] |
| $\Delta t$ | Ultrasonic echo pulse duration round-trip time [$\mu$s] |
| $c(T)$ | Speed of sound in ambient air at temperature $T$ [m/s] |
| $s_0$ | Minimum jam distance in Intelligent Driver Model [m] |
| $T$ | Desired safe headway time gap in IDM [s] |

*Dedicated to my parents, mentors, and the urban cycling community striving for safer streets.*

CHAPTER 1

# Introduction

## 1.1 Background and Context

Urban transportation systems worldwide are experiencing an imperative modal transition toward sustainable, non-motorized micro-mobility, centered predominantly on utility and commuter bicycling. Bicycling delivers well-documented societal dividends, including zero-emission urban transit, mitigation of localized traffic congestion, lower road infrastructure degradation, and substantial cardiovascular health improvements for citizens. Consequently, metropolitan transit authorities have prioritized cycling promotion through active mobility charters, expanded cycle path networks, and public bicycle sharing programs.

However, the widespread adoption of utility cycling remains severely constrained by objective physical dangers and subjective risk perceptions. Cyclists operate as Vulnerable Road Users (VRUs), possessing minimal physical protection compared to occupants of passenger cars, trucks, and buses. In mixed-traffic corridors where dedicated, physically separated cycling infrastructure is absent, cyclists share the right-of-way directly with motorized vehicles characterized by significantly higher operating velocities, larger kinetic masses, and extensive blind spots. Under these conditions, motorized overtaking maneuvers represent one of the most hazardous interaction regimes in urban traffic dynamics.

## 1.2 Problem Statement

When a motorized vehicle overtakes a bicycle in a shared lane or an advisory cycle lane, insufficient lateral clearance—commonly designated as a “close pass”—exposes the cyclist to extreme collision hazards, aerodynamic buffeting turbulence, startle-induced steering instabilities, and severe subjective trauma. Despite the severity of this hazard, traffic authorities and safety engineers face two critical methodological deficiencies:

1. **Under-Reporting and Retrospective Lag of Incident Records:** Municipal road safety planning relies almost exclusively on retrospective police accident registries. However, near-miss close passes, aggressive overtakes without contact, and emergency avoidance swerves rarely appear in official statistics unless catastrophic injury or property damage occurs. As a result, hazardous road segments remain officially unflagged for years.

2. **Lack of Objective Telemetry and Context-Aware Conflict Classification:** Existing naturalistic cycling studies often rely on crude smartphone GPS loggers or manual

rider questionnaires. Smartphone crowd-sensing frameworks (such as SimRA) record inertial shocks but lack lateral acoustic transducers to measure physical passing distances. Conversely, commercial radar sensors produce binary proximity flags without logging vehicle class distinctions (e.g., distinguishing between a 1.2 m pass by a peer bicycle versus a 1.2 m pass by a 15-ton multi-axle freight truck) or compensating for the bicycle's dynamic roll angle when leaning into turns.

## 1.3 Motivation and Significance

To curb overtaking hazards, modern traffic regulations have instituted strict statutory clearance mandates. Most prominently, the German Road Traffic Act (*Straßenverkehrs-Ordnung*, StVO) § 5 Abs. 4 was amended in 2020 to mandate a minimum lateral passing clearance of **1.50 meters** within built-up urban boundaries and **2.00 meters** outside urban limits [1]. Similar minimum passing distance laws (MPDL) have been enacted across multiple jurisdictions internationally [2].

Enforcing and evaluating these spatial regulations requires objective, empirical measurement infrastructure. An instrumented telemetry device capable of continuously logging centimeter-accurate lateral clearances, GPS coordinates, and three-axis inertial dynamics can map city-wide overtaking risk profiles. Furthermore, because physical sensor deployment across thousands of commuter bicycles is resource-intensive and logistically challenging, coupling physical IoT hardware telemetry with high-fidelity **microscopic traffic simulation** creates a powerful digital twin. Using calibrated microscopic simulators such as Eclipse SUMO [3], traffic researchers can model full-city traffic flows, test infrastructure redesigns, and predict cyclist conflict rates under varying traffic volumes before investing millions in civil road construction.

## 1.4 Aim and Objectives

The overall **aim** of this project is to develop **CycleSafe**, an integrated open-architecture platform that combines instrumented bicycle hardware telemetry, microscopic multi-modal traffic simulation, and machine learning conflict classification to quantitatively analyze and mitigate urban cyclist overtaking risks.

To achieve this primary aim, five concrete and measurable **objectives** are formulated:

1. **Objective 1 (Embedded Hardware Architecture):** Design and engineer an open-hardware telemetry sensor node based on the ESP32 microcontroller, dual waterproof ultrasonic transducers (JSN-SR04T), satellite positioning (u-blox NEO-6M), and an MPU-6050 6-axis IMU, incorporating an analytical roll-angle tilt compensation model ($\phi = \arctan(a_y/\sqrt{a_x^2 + a_z^2})$) and a 3.3 V power regulation rail to ensure non-volatile SD telemetry logging.
2. **Objective 2 (Mechanical Packaging & Enclosure):** Synthesize and package the production-ready 3D printable mechanical housing in PETG material, providing weatherproof acoustic sensor brackets, handlebar display mounts, and universal seatpost clamping.

3. **Objective 3 (Microscopic Simulation Digital Twin):** Construct a calibrated, 10 km multi-lane corridor simulation in Eclipse SUMO along Berlin's canonical thoroughfare (Ernst-Reuter-Platz to Oberbaumbrücke), driven by empirical 12-hour vehicular census data (13,531 motor vehicles and 1,397 bicycles) normalized to a one-hour stochastic traffic population.
4. **Objective 4 (Machine Learning Conflict Classification):** Train and integrate an Extreme Gradient Boosting (XGBoost) classifier on the TU Berlin SimRA empirical dataset to evaluate overtaking encounters across three discrete severity tiers (*Safe Pass*, *Close Pass (Medium)*, and *Severe Incident*), incorporating German StVO § 5(4) distance metrics and counterpart vehicle mass penalties.
5. **Objective 5 (Full-Stack Ingestion and Web Visualization):** Develop a responsive, containerized web platform (Go, PostgreSQL, Redis, Leaflet, Chart.js) featuring multi-bicycle trajectory inspection, spatial conflict heatmaps, kinematic acceleration graphs, and one-click telemetry CSV export.

## 1.5 Scope and Limitations

The scope of this project encompasses the end-to-end design, software implementation, circuit architecture, and simulation validation of the CycleSafe platform:

- **Spatial Scope:** The microscopic simulation is calibrated on a 9.53 km representative arterial corridor in Berlin connecting Charlottenburg (Point A: Ernst-Reuter-Platz) to Friedrichshain-Kreuzberg (Point B: Oberbaumbrücke), representing mixed urban lanes, segregated cycle paths, and signalized junctions.
- **Empirical Calibration Scope:** Traffic demand volumes and modal splits are derived directly from the official 2024/2025 Berlin Senate traffic census (*Senatsverwaltung für Mobilität, Verkehr, Klimaschutz und Umwelt*) [4].
- **Hardware Prototyping Scope:** Full electrical schematics, pinout mapping, component procurement verification (Indian domestic suppliers delivering to Dhanbad PIN 826004), 3D STL enclosure packaging, and mathematical calibration are completed. Physical road testing is designed for subsequent campus deployment.
- **Exclusions:** The simulation model utilizes a controlled multi-lane dual-corridor parameterized with the Intelligent Driver Model (IDM) and TraCI lateral sub-lane dynamics, rather than an uncalibrated 100,000-edge city-wide raw OSM mesh, ensuring 100% deterministic collision tracking without junction deadlocks.

## 1.6 Contributions

The primary contributions of this project are summarized as follows:

1. **Inertial Tilt-Compensated Sensing Model:** Formulation and mathematical derivation of an MPU-6050 dynamic roll compensation equation that adjusts raw acoustic

Time-of-Flight measurements ($d_{\text{true}} = d_{\text{measured}} \cos \phi$) to correct for cyclist cornering lean and pedaling sway.

2. **Microscopic TraCI Multi-Vehicle Traffic Engine:** An automated SUMO simulation pipeline generating stochastic multi-agent interactions across 1,244 concurrent vehicles with configurable random seeds and real-time kinematic telemetry extraction.

3. **Vehicle-Class Aware XGBoost Safety Engine:** A supervised classifier replacing arbitrary static distance thresholds with empirical near-miss criteria that account for speed differentials, deceleration rates, and vehicle mass disparities.

4. **Full-Stack Open Architecture:** An open-source, containerized web platform providing asynchronous simulation dispatching, multi-bicycle ride switching, and CSV telemetry exports.

## 1.7 Organization of the Report

The remainder of this report is organized as follows:

- **Chapter 2: Literature Review and Related Work** provides a comprehensive survey of proximity sensing technologies, crowdsourced cycling safety initiatives (SimRA, OpenBikeSensor), and microscopic traffic simulation benchmarks.

- **Chapter 3: Requirements and Problem Formulation** outlines the application scenarios, functional and non-functional requirements, and the formal mathematical definitions of overtaking clearance.

- **Chapter 4: Proposed Methodology and System Design** details the platform architecture, acoustic physics, roll tilt math, XGBoost model formulation, and database schema.

- **Chapter 5: Implementation and Experimental Setup** documents the embedded hardware bill of materials, circuit pinouts, 3D printing parameters, and SUMO simulation parameters.

- **Chapter 6: Results and Discussion** presents the experimental validation, quantitative conflict distributions across vehicle modes, ML classification accuracy, and sensitivity analyses.

- **Chapter 7: Conclusion and Future Work** synthesizes the principal findings, evaluates objective fulfillment, and outlines a technical roadmap for field deployment.

# CHAPTER 2

# Literature Review and Related Work

## 2.1 Review Approach

To establish a rigorous theoretical and empirical foundation for the CycleSafe platform, a systematic literature survey was conducted across peer-reviewed transportation science journals, intelligent transportation systems (ITS) conferences, and open-source hardware repositories. The review focused on four primary literature thematic clusters published between 2010 and 2026:

1. Naturalistic cycling studies measuring physical motor vehicle overtaking distances.
2. Embedded IoT proximity sensing nodes and hardware telemetry units deployed on non-motorized vehicles.
3. Smartphone-based crowdsourcing and inertial near-miss detection platforms.
4. Microscopic multi-modal traffic simulation frameworks modeling bicycle-vehicle interactions.

## 2.2 Thematic Review

### 2.2.1 Naturalistic Overtaking Studies and Passing Distance Laws

Over the past decade, empirical naturalistic cycling studies have established that motorized overtaking clearance directly governs cyclist perceived safety and objective injury risk [5, 6]. Early pioneering research by Walker utilized instrumented bicycles equipped with ultrasonic rangefinders and video cameras to quantify driver behavior, revealing that vehicle speed, oncoming traffic density, and lane width significantly constrict overtaking margins.

Following the global proliferation of Minimum Passing Distance Laws (MPDL)—exemplified by Germany's StVO § 5(4) mandating a 1.5 m clearance in urban zones [1] and Australia's 1.0 m/1.5 m statutory rules [2]—researchers turned to dedicated telemetry devices to assess compliance. Beck et al. [2] demonstrated in a multi-city Australian naturalistic cycling cohort that between 10% and 19% of all overtaking events violated statutory passing distance laws, with higher violation rates occurring on roads without marked bicycle lanes and along curved segments.

### 2.2.2 Embedded Hardware Sensing: The OpenBikeSensor (OBS) Platform

The most prominent open-hardware initiative targeting passing distance measurement is the **OpenBikeSensor (OBS)** project [7]. Developed by civic technology communities in Germany, OBS employs an Espressif ESP32 dual-core microcontroller interfaced with two lateral waterproof ultrasonic transducers (JSN-SR04T), a u-blox satellite receiver, a small OLED display, and a handlebar momentary push-button.

In the OBS workflow, ultrasonic pulses continuously measure lateral clearance to the left (overtaking motorized vehicles) and to the right (parked vehicles and roadside infrastructure). When an overtaking event occurs, the rider actuates the handlebar button, causing the firmware to record a confirmation marker in the telemetry log. While OBS represents a milestone in citizen science and community data collection, several critical engineering limitations remain unaddressed in the literature:

- **Absence of Inertial Tilt Compensation:** Bicycles lean into turns and oscillate laterally during active pedaling. When a bicycle tilts at a roll angle $\phi$, an ultrasonic transducer rigidly attached to the frame emits an acoustic cone directed obliquely downward toward the pavement or upward toward the vehicle roofline. Without real-time accelerometer roll compensation, this geometric tilt introduces systematic distance measurement errors.
- **Power Rail Instability and SD Write Corruption:** Operating the ESP32 Wi-Fi radio, ultrasonic transducers, and MicroSD card SPI bus from unbuffered battery rails frequently induces transient brownout resets, causing FAT32 file allocation table corruptions during long recording sessions.

### 2.2.3 Crowdsourced Smartphone Incident Sensing: The SimRA Platform

As an alternative to specialized hardware sensors, researchers at Technische Universität Berlin developed **SimRA** (Safety in Mobility: Ride Analytics) [8]. SimRA utilizes the embedded tri-axial accelerometers and GPS receivers of consumer smartphones mounted on bicycle handlebars to passively detect sudden braking ($\Delta a_x < -2.5\,\text{m/s}^2$) and sharp swerving maneuvers ($\Delta a_y$). At the conclusion of a ride, the cyclist annotates flagged near-miss incidents with contextual details (e.g., cut-ins, car door openings, close passes).

While SimRA successfully engaged tens of thousands of participants and collected millions of kilometers of urban bicycle trajectory data across European cities, its reliance on smartphones incurs fundamental physical limitations:

1. Smartphones possess no active ranging sensors (ultrasonic or radar); consequently, passing distance cannot be measured physically and relies entirely on subjective post-ride estimation.
2. Smartphone mounting orientations vary widely, and mechanical phone holders introduce high-frequency vibrational noise that complicates deterministic acceleration thresholding.

### 2.2.4 Microscopic Multi-Modal Traffic Simulation in SUMO

Microscopic traffic simulation offers a reproducible computational laboratory for modeling complex vehicular interactions without the logistical overhead and safety risks of on-road trials. **Eclipse SUMO (Simulation of Urban MObility)** [3, 9] is the leading open-source microscopic simulation suite. SUMO implements car-following models such as the Intelligent Driver Model (IDM) [10] and the Krauß model, alongside the Sublane Model, which divides discrete road lanes into continuous lateral strips (e.g., 0.2 m resolution) to simulate realistic non-lane-based micro-mobility overtaking.

Through the Traffic Control Interface (TraCI), external control routines can subscribe to moving vehicle variables (spatial coordinates, instantaneous speeds, acceleration vectors, and contextual distance envelopes) at each simulation step. However, existing SUMO literature predominantly focuses on motorized vehicle delay minimization and intersection throughput, rarely modeling the continuous lateral acoustic clearance and inertial roll dynamics experienced by individual cyclists in mixed-traffic streams.

## 2.3 Comparative Analysis of Related Approaches

To contextualize the CycleSafe contribution within the current technological landscape, Table 2.1 presents a comprehensive comparative benchmark against existing platforms.

## 2.4 Identified Research Gaps

The literature survey identifies three primary research gaps that directly motivate this investigation:

1. **Gap 1: Geometric Tilt Distortion in Bicycle Ranging Telemetry.** Existing hardware proximity platforms (including OpenBikeSensor and ProxiCycle) assume the instrumented bicycle operates perpetually in an upright perpendicular posture. In real-world urban cycling, cornering, maneuvering around road obstacles, and standard pedaling cadence produce dynamic roll angles between $5^\circ$ and $25^\circ$, introducing systematic distance overestimation that invalidates close-pass statistical baselines.

2. **Gap 2: Uniform Severity Treatment Across Disparate Vehicle Classes.** Conventional threshold-based collision warning systems treat all lateral clearances below 1.5 m identically. However, in urban freight corridors, an overtaking heavy-duty truck passing at 1.4 m presents vastly greater turbulence, kinetic mass threat, and cognitive panic than a peer bicycle passing at 1.4 m on a separated path. There is a lack of machine learning models that fuse physical clearance with counterpart vehicle taxonomy.

3. **Gap 3: Decoupling of Empirical IoT Logging and Microscopic Simulation.** Naturalistic IoT telemetry collection and microscopic simulation modeling have progressed along isolated tracks. Physical sensor data is rarely used to parameterize and

Table 2.1: Comparative benchmarking of cycling safety evaluation platforms.

| **Criteria** | **ProxiCycle** | **SimRA** | **OpenBikeSensor** | **CycleSafe (Ours)** |
|---|---|---|---|---|
| **Core Hardware** | Arduino / RPi | Smartphone | ESP32 Dual-Core | ESP32 + MPU-6050 |
| **Ranging Sensors** | IR / Ultrasonic | None (Inertial) | Dual JSN-SR04T | Dual JSN-SR04T |
| **Distance Output** | Binary flag | Subjective survey | Continuous (cm) | Continuous (cm) |
| **Tilt Compensation** | None | None | None | Roll Model ($\phi = \arctan \frac{a_y}{\sqrt{a_x^2+a_z^2}}$) |
| **Vehicle Taxonomy** | None | Post-ride survey | Generic acoustic | XGBoost Classifier |
| **Simulation Twin** | None | None | None | 10 km SUMO TraCI (1,244 Veh) |
| **Power Reliability** | Unregulated | Internal battery | Shared rail | Dedicated AMS1117 LDO |
| **Telemetry Export** | None | KML / CSV | Web JSON | REST API + Instant CSV |

validate multi-modal simulation twins, and traffic simulation platforms rarely generate synthetic telemetry matching the exact schema and physics of embedded hardware nodes.

## 2.5 Chapter Summary

This chapter reviewed the state of the art in cyclist overtaking safety evaluation, spanning statutory passing regulations, embedded telemetry hardware, smartphone crowdsourcing, and microscopic traffic simulation. The critical shortcomings of existing solutions—lack of dynamic roll tilt compensation, absence of vehicle-specific hazard weighting, and the methodological divide between physical sensors and simulation models—form the explicit research rationale for the CycleSafe architecture presented in subsequent chapters.

# CHAPTER 3

# Requirements and Problem Formulation

## 3.1 Application Scenario

The operational environment for CycleSafe comprises dense metropolitan road networks characterized by mixed traffic flows, variable infrastructure typologies, and high vehicle density. The primary application workflow proceeds across two interconnected operational domains:

1. **Physical Sensing Scenario:** An urban utility or commuter cyclist mounts the CycleSafe telemetry unit on their bicycle (main sensor pod on the seatpost beneath the saddle, and remote display/trigger pod on the handlebar). As the cyclist traverses their daily commute, the device continuously emits ultrasonic ranging pulses to monitor lateral clearance to passing traffic and roadside obstacles at 20 Hz, samples geographic coordinates at 1 Hz via GPS, and monitors 6-axis inertial motion at 100 Hz. When an overtaking vehicle performs a dangerous maneuver, the cyclist presses the handlebar button. All telemetry is stored on an onboard FAT32 MicroSD card.
2. **Microscopic Simulation Scenario:** Urban transport planners, municipal engineers, and researchers access the CycleSafe web platform. Rather than waiting months to collect empirical rides across every city block, the user selects an urban corridor (e.g., Berlin's 10 km Ernst-Reuter-Platz to Oberbaumbrücke corridor) and specifies a traffic demand window (e.g., peak morning rush hour 07:00–08:00). The simulation engine spins up a microscopic multi-agent digital twin simulating 1,244 moving vehicles, executes the XGBoost ML conflict evaluator, and generates telemetry streams identical in structure to the physical hardware node.

## 3.2 Functional Requirements

The functional capabilities of the CycleSafe platform are codified into eight formal requirements:

- **FR-1: Dual-Sided Acoustic Ranging:** The embedded system shall measure lateral distances independently on both left and right flanks over a measurement span of 0.20 m to 4.50 m with a millimeter-level resolving capability.

- **FR-2: Dynamic Roll-Angle Tilt Compensation:** The system shall continuously measure tri-axial acceleration, compute the instantaneous frame roll angle $\phi$, and apply trigonometric correction to raw acoustic distances in real time.
- **FR-3: Geospatial Georeferencing:** The device shall acquire and log WGS84 latitude and longitude coordinates with active HDOP monitoring and NMEA-0183 checksum validation.
- **FR-4: User Event Triggering:** The handlebar unit shall register manual rider confirmations via an interrupt-driven momentary switch, recording a timestamped event flag correlated with ultrasonic and IMU vectors.
- **FR-5: Fail-Safe Local Persistence:** Sensor readings shall be serialized and flushed to non-volatile MicroSD storage in CSV format, protected against brownout file system corruption.
- **FR-6: Microscopic Multi-Modal Traffic Simulation:** The simulation subsystem shall execute 60-minute multi-vehicle traffic runs in Eclipse SUMO utilizing the sublane model (lateral resolution $\leq 0.2\,\text{m}$) and the Intelligent Driver Model (IDM).
- **FR-7: Multi-Class Machine Learning Conflict Classification:** The platform shall classify every vehicular interaction into one of three discrete categories (*Safe Pass*, *Close Pass (Medium)*, or *Severe Incident*) using an empirical XGBoost model that accounts for clearance distance, relative speed, acceleration, and counterpart vehicle category.
- **FR-8: Interactive Spatial Visualization and Telemetry Export:** The web interface shall visualize simulated and physical rides on interactive Leaflet maps, render kinematic time-series charts, and allow one-click export of complete trajectory points to CSV.

## 3.3 Non-Functional Requirements

To ensure practical field utility and academic reliability, the platform satisfies the following non-functional criteria:

- **NFR-1 (Ranging Accuracy):** Lateral distance measurements shall maintain a maximum absolute error of $\leq \pm 2\,\text{cm}$ within the critical 0.5 m to 2.5 m overtaking zone under ambient operating temperatures ($0°\text{C}$ to $45°\text{C}$).
- **NFR-2 (Latency and Real-Time Responsiveness):** The embedded firmware loop shall execute ultrasonic trigger-echo sampling, IMU polling, and OLED display refresh within a strict latency bound of $< 50\,\text{ms}$ per cycle.
- **NFR-3 (Operational Battery Autonomy):** The embedded hardware shall operate continuously for a minimum of 6.0 hours on a single 18650 2500 mAh lithium-ion cell under full sensor polling and active logging.

- **NFR-4 (Weather and Mechanical Durability):** The physical enclosure shall exhibit IP65 splash resistance, utilizing PETG 3D-printed housing capable of withstanding solar thermal exposure up to $80°\text{C}$ without geometric softening.
- **NFR-5 (Simulation Determinism and Reproducibility):** The microscopic simulation engine shall accept arbitrary pseudo-random seeds to enable either perfectly deterministic regression testing or stochastic variance across simulation runs.
- **NFR-6 (Low Component Cost):** The complete bill of materials for the physical telemetry node shall not exceed INR 4,000 ( 45 USD), sourced entirely through domestic Indian electronics vendors with active shipping to IIT (ISM) Dhanbad (PIN: 826004).

## 3.4 Constraints and Assumptions

### 3.4.1 Hardware and Operational Constraints

1. **Ultrasonic Acoustic Blind Zone:** The JSN-SR04T transducer exhibits a physical acoustic ring-down blind zone of 20 cm. Objects closer than 0.20 m from the transducer faceplate cannot be resolved acoustically.
2. **I2C Bus Address Constraints:** The Solomon Systech SSD1306 OLED controller is hardwired to 7-bit slave address `0x3C`, while the InvenSense MPU-6050 is configured to address `0x68` by pulling pin `AD0` to ground.
3. **Single-Cell Lithium Voltage Characteristics:** A standard 18650 NMC battery operates across a dynamic voltage curve from 4.2 V (full charge) down to 3.0 V (cutoff). Because the ESP32 and MicroSD card require a stable 3.3 V rail, a low-dropout regulator with dropout voltage $V_{\text{drop}} < 200\,\text{mV}$ at typical operating currents is mandatory.

### 3.4.2 Simulation Modeling Assumptions

1. **Corridor Boundary Conditions:** In the 10 km simulation corridor, all generated vehicles depart at the western boundary (Ernst-Reuter-Platz) and exit at the eastern boundary (Oberbaumbrücke), representing continuous arterial traffic.
2. **Vehicle Modal Homogeneity:** Within each vehicle class (passenger cars, trucks, motorcycles, bicycles), physical dimensions and maximum braking decelerations follow standard German federal fleet parameters defined in the SUMO vehicle type definitions.

## 3.5 Formal Problem Definition

Let the state of the instrumented ego bicycle at discrete simulation timestamp $t$ be defined by the tuple:

$$S_{\text{ego}}(t) = (\mathbf{p}_{\text{ego}}(t), v_{\text{ego}}(t), a_{\text{ego}}(t), \phi(t)) \tag{3.1}$$

where $\mathbf{p}_{\text{ego}}(t) = (\lambda_{\text{ego}}(t), \mu_{\text{ego}}(t))$ denotes the WGS84 geographic coordinate (longitude, latitude), $v_{\text{ego}}(t)$ is the scalar longitudinal velocity, $a_{\text{ego}}(t)$ is the acceleration vector, and $\phi(t)$ is the frame roll tilt angle.

Similarly, let a motorized counterpart vehicle $k \in \{1, 2, \ldots, K\}$ operating in spatial proximity be represented by:

$$S_k(t) = (\mathbf{p}_k(t), v_k(t), a_k(t), C_k, L_k, W_k) \tag{3.2}$$

where $C_k \in \{\text{car}, \text{truck}, \text{motorbike}, \text{bike}\}$ denotes the counterpart vehicle class, and $L_k, W_k$ represent bounding length and width dimensions.

### 3.5.1 Edge-to-Edge Spatial Clearance

The true physical clearance $d_{\text{clear}}(t, k)$ between the lateral bounding edge of the bicycle and the nearest lateral surface of the counterpart vehicle is formulated as:

$$d_{\text{clear}}(t, k) = \max\left(0,\ \|\mathbf{p}_k(t) - \mathbf{p}_{\text{ego}}(t)\| - \frac{W_{\text{ego}} + W_k}{2}\right) \tag{3.3}$$

In the physical hardware telemetry unit, this clearance is measured by the lateral ultrasonic transducer and corrected for roll angle:

$$d_{\text{true}}(t) = d_{\text{measured}}(t) \cdot \cos\left(\phi(t)\right) \tag{3.4}$$

### 3.5.2 Kinematic Conflict Metrics

To assess encounter criticality, two classical microscopic conflict metrics are formulated:

1. **Time-to-Collision (TTC):** When the counterpart vehicle approaches the bicycle from behind in the same or adjacent lane with relative closing velocity $\Delta v(t) = v_k(t) - v_{\text{ego}}(t) > 0$:
$$\text{TTC}(t, k) = \frac{\Delta x_{\text{long}}(t, k)}{\Delta v(t)} = \frac{x_k(t) - x_{\text{ego}}(t)}{v_{\text{ego}}(t) - v_k(t)} \tag{3.5}$$

2. **Post-Encroachment Time (PET):** The temporal interval between the departure of the bicycle from a shared lane conflict zone and the arrival of the overtaking vehicle at the exact same spatial coordinates:
$$\text{PET} = t_{\text{arrival}}(k, \mathbf{p}^*) - t_{\text{departure}}(\text{ego}, \mathbf{p}^*) \tag{3.6}$$

### 3.5.3 Multi-Class Hazard Optimization

The classification objective is to map the multidimensional encounter observation vector $\mathbf{x} = [d_{\text{clear}},\, \Delta v,\, a_{\text{ego}},\, a_k,\, C_k]$ into a discrete severity label $y \in \{0, 1, 2\}$:

$$y = \begin{cases} 0 & \text{(Safe Pass: StVO compliance, } d_{\text{clear}} \geq 1.5\,\text{m)} \\ 1 & \text{(Close Pass Medium: } 1.0\,\text{m} \leq d_{\text{clear}} < 1.5\,\text{m or mild deceleration)} \\ 2 & \text{(Severe Incident: } d_{\text{clear}} < 1.0\,\text{m or emergency braking } a_{\text{ego}} \leq -2.5\,\text{m/s}^2\text{)} \end{cases} \tag{3.7}$$

This formulation underpins the machine learning classification engine detailed in the following chapter.

CHAPTER 4

# Proposed Methodology and System Design

## 4.1 Overall Architecture

The CycleSafe platform is structured as a dual-layer cyber-physical architecture comprising two complementary operational subsystems:

1. **Embedded Physical Telemetry Node (Edge Layer):** An instrumented on-bicycle IoT sensing node that captures real-world lateral ultrasonic clearances, geographic coordinates, 6-axis inertial dynamics, and manual cyclist event confirmations.
2. **Microscopic Simulation & ML Analytics Engine (Cloud/Server Layer):** A containerized backend service running Eclipse SUMO, TraCI, PostgreSQL, Redis, and a supervised XGBoost classification model, exposed via a full-stack web application.

Figure 4.1 visualizes the cross-layer system dataflow and component interconnections.

## 4.2 Design Rationale

The selection of hardware components and software libraries is guided by strict engineering tradeoffs between measurement accuracy, power consumption, cost, and open-source accessibility:

- **Espressif ESP32 Microcontroller:** Chosen for its dual-core 32-bit architecture operating at 240 MHz with 520 KB SRAM. Core 0 is dedicated to high-frequency FreeRTOS sensor sampling (ultrasonic pulse capture and IMU polling), while Core 1 handles NMEA sentence parsing, OLED display rendering, and SPI MicroSD block writes without timing jitter.
- **JSN-SR04T Waterproof Transducers:** Unlike exposed open-mesh ultrasonic sensors (such as HC-SR04), the JSN-SR04T features an enclosed, sealed aluminum transducer probe connected via a 2.5 m coaxial cable. This design provides IP67 environmental protection against rain, road splashes, and dust, essential for outdoor cycling.

| **LAYER 1: EMBEDDED PHYSICAL TELEMETRY NODE (Edge Hardware)** | | |
|---|---|---|
| **Ranging & Kinematics** | **Microcontroller & Compute** | **Power & User Interface** |
| Dual JSN-SR04T (20 Hz) | ESP32 Dual-Core (240 MHz) | 18650 Li-ion 2500 mAh (3.7 V) |
| MPU-6050 6-Axis IMU | FreeRTOS Dual-Task System | TP4056 Type-C Charger |
| u-blox NEO-6M GPS | FAT32 MicroSD SPI Logger | AMS1117-3.3V LDO Rail |
| Handlebar Button (12 mm) | I2C Bus (GPIO 21/22) | 0.96” SSD1306 OLED |
| ⇓ *Standardized Telemetry CSV Schema* $\{t$, lat, lon, $d_L$, $d_R$, $a_x$, $a_y$, $a_z$, btn$\}$ ⇓ | | |
| **LAYER 2: MICROSCOPIC SIMULATION & ML PLATFORM (Server)** | | |
| **Simulation Subsystem** | **Machine Learning Engine** | **Web Application & Storage** |
| Eclipse SUMO 1.27.1 + TraCI | XGBoost 3-Class Booster | Go 1.23 REST API Service |
| 10 km Berlin Arterial Corridor | Trained on SimRA Dataset | PostgreSQL 16 Relational DB |
| 1,244 Stochastic Vehicles | StVO § 5(4) Rule Encoders | Redis 7 In-Memory Cache |
| Sublane Lateral Dynamics (0.2 m) | Modal Severity Matrix | Leaflet + Chart.js Interface |

Figure 4.1: End-to-end architectural dataflow of the CycleSafe platform.

- **Shared I2C Bus Multiplexing:** Both the 0.96” SSD1306 OLED display and the MPU-6050 IMU communicate over the identical physical two-wire interface (`GPIO 21 = SDA`, `GPIO 22 = SCL`). Because the OLED controller resides at 7-bit address `0x3C` and the MPU-6050 resides at `0x68`, the bus supports concurrent operation with zero additional GPIO pin overhead.
- **Dedicated AMS1117-3.3V Power Regulation Rail:** During write operations, MicroSD cards draw transient current spikes exceeding 100 mA, while the ESP32 RF subsystem draws up to 240 mA during Wi-Fi beacon transmission. Supplying these peripherals from a dedicated low-dropout regulator isolated from the raw battery prevents voltage sagging below the 2.7 V brownout threshold.

## 4.3 Algorithms and Mathematical Models

### 4.3.1 Acoustic Time-of-Flight Ranging Physics

The physical distance $d_{\text{measured}}$ from the ultrasonic transducer faceplate to the nearest reflecting vehicle surface is governed by the two-way acoustic Time-of-Flight equation:

$$d_{\text{measured}} = \frac{c(T) \cdot \Delta t}{2} \tag{4.1}$$

where $\Delta t$ is the measured echo pulse duration (round-trip transit time in microseconds), and $c(T)$ is the acoustic propagation velocity in air. The speed of sound is sensitive to ambient air temperature $T$ (°C):

$$c(T) = 331.3 \cdot \sqrt{1 + \frac{T}{273.15}} \approx 331.3 + 0.606 \cdot T \quad [\text{m/s}] \tag{4.2}$$

At standard ambient temperature (20°C), $c(20) = 343.4$ m/s, yielding a spatial resolution of $0.1717$ mm per microsecond of echo pulse duration.

### 4.3.2 Analytical Dynamic Roll-Angle Tilt Compensation

During cornering and pedaling, the bicycle frame tilts at an instantaneous roll angle $\phi$ relative to the true gravitational vertical. An uncompensated transducer rigidly mounted to the frame projects an inclined acoustic beam, measuring an oblique hypotenuse distance $d_{\text{measured}}$ rather than the true horizontal ground clearance $d_{\text{true}}$ (Figure 4.2).

**Dynamic Roll-Angle Tilt Compensation Model**

| | |
|---|---|
| Gravitational Vector: | $\mathbf{g} = [0, 0, -9.81]^T$ |
| Measured Acceleration: | $\mathbf{a} = [a_x, a_y, a_z]^T$ (from MPU-6050) |
| Roll Angle Formulation: | $\phi = \arctan\left(\frac{a_y}{\sqrt{a_x^2 + a_z^2}}\right)$ |
| Corrected Horizontal Clearance: | $d_{\text{true}} = d_{\text{measured}} \cdot \cos(\phi)$ |

Figure 4.2: Formulation of dynamic roll-angle tilt compensation.

The MPU-6050 IMU samples three-axis acceleration vectors $(a_x, a_y, a_z)$ at 100 Hz. The instantaneous roll angle $\phi$ is calculated analytically using:

$$\phi = \arctan\left(\frac{a_y}{\sqrt{a_x^2 + a_z^2}}\right) \tag{4.3}$$

The true perpendicular ground clearance to the overtaking vehicle is then computed by projecting the measured hypotenuse onto the horizontal plane:

$$d_{\text{true}} = d_{\text{measured}} \cdot \cos(\phi) \tag{4.4}$$

For example, if a cyclist leans into a corner at $\phi = 20°$ and the sensor records $d_{\text{measured}} = 1.60\,\text{m}$, the uncompensated reading would classify the overtake as safe ($> 1.5\,\text{m}$), whereas the true horizontal distance is $d_{\text{true}} = 1.60 \cdot \cos(20°) = 1.503\,\text{m}$, near the critical threshold.

### 4.3.3 XGBoost Supervised Conflict Classification Engine

Rather than applying a rigid, one-size-fits-all threshold of 1.5 m regardless of context, CycleSafe employs an Extreme Gradient Boosting (**XGBoost**) [11] multi-class classifier. The feature vector $\mathbf{x}_i \in \mathbb{R}^6$ extracted for each vehicular encounter comprises:

$$\mathbf{x}_i = [d_{\text{min}},\ v_{\text{ego}},\ v_{\text{other}},\ a_{\text{ego,min}},\ \Delta v,\ C_{\text{type}}] \tag{4.5}$$

where:

- $d_{\text{min}}$ is the minimum roll-compensated clearance observed during the encounter [m],
- $v_{\text{ego}}, v_{\text{other}}$ are the instantaneous velocities of bicycle and vehicle [m/s],
- $a_{\text{ego,min}}$ is the peak longitudinal deceleration recorded by the cyclist [m/s$^2$],
- $\Delta v = v_{\text{other}} - v_{\text{ego}}$ is the relative overtaking velocity [m/s],
- $C_{\text{type}} \in \{0 : \text{bike}, 1 : \text{car}, 2 : \text{truck/bus}, 3 : \text{motorbike}\}$ is the integer-encoded vehicle class.

The XGBoost model minimizes a regularized objective function across $K$ additive decision trees:

$$\mathcal{L}(\Theta) = \sum_{i=1}^{N} l\left(y_i, \hat{y}_i\right) + \sum_{k=1}^{K} \Omega(f_k) \tag{4.6}$$

where the loss function $l$ is multi-class cross-entropy, and the tree complexity regularization term $\Omega(f_k)$ penalizes leaf count $T$ and weight vector $\mathbf{w}$:

$$\Omega(f_k) = \gamma T + \frac{1}{2}\lambda \sum_{j=1}^{T} w_j^2 \tag{4.7}$$

The optimal split at each node is determined by maximizing the split gain criterion:

$$\text{Gain} = \frac{1}{2}\left[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda}\right] - \gamma \tag{4.8}$$

where $G$ and $H$ represent first and second-order gradient statistics of the loss function.

### 4.3.4 Encoding German StVO Statutory Clearances and Modal Penalties

The classifier integrates the legal requirements of German StVO § 5(4) [1] by enforcing asymmetric loss penalties. Passing clearances below 1.5 m are inherently penalized. Furthermore, counterpart vehicle mass penalties are applied: an overtaking truck or municipal bus ($C_{\text{type}} = 2$) displacing significant aerodynamic volume and possessing blind zones requires a widened safety buffer ($d_{\text{safe}} \geq 2.0\,\text{m}$), whereas peer bicycles passing on segregated paths ($C_{\text{type}} = 0$) are considered non-conflicting down to clearances of 0.80 m.

## 4.4 Data and Empirical Sources

The models and simulations in CycleSafe are calibrated using two empirical open-access transportation datasets:

1. **TU Berlin SimRA Open Dataset:** Comprises over 25,000 real-world bicycle rides collected across Berlin [8]. Ground-truth near-miss annotations and accelerometer traces were extracted to train and validate the XGBoost classifier.
2. **Berlin Senate Traffic Census 2024/2025 (Oberbaumbrücke Counting Station):** Official 12-hour automated traffic counts (07:00–19:00) provided by the *Senatsverwaltung für Mobilität, Verkehr, Klimaschutz und Umwelt* (Document S19-23905 and *Bericht Radverkehr 2025*) [4]. The dataset records:
    - Passenger Cars and Light Commercial Vehicles (Pkw + Lfw): 12,260 (90.6% of motor vehicles)
    - Heavy-Duty Trucks and Buses (Lkw + Bus): 676 (5.0%)
    - Motorbikes and Scooters (Krad): 595 (4.4%)
    - Commuter Bicycles: 1,397

## 4.5 Ethics, Privacy, and Data Governance

Because cycling telemetry captures fine-grained spatiotemporal trajectories that could reveal residential addresses or personal daily routines, CycleSafe incorporates privacy-by-design principles aligned with EU General Data Protection Regulation (GDPR) standards:

- **No Optical Surveillance:** The telemetry unit relies entirely on ultrasonic time-of-flight acoustic pulses and inertial sensors; no video cameras or facial recognition systems are deployed.
- **Spatial Obfuscation at Trip Terminals:** In production ingestion, the initial and concluding 200 meters of recorded GPS tracks are clipped to prevent home or workplace deanonymization.

# CHAPTER 5

# Implementation and Experimental Setup

## 5.1 Development Environment

The CycleSafe computational environment is designed for containerized reproducibility across Unix and Windows hosts. The software stack is orchestrated via Docker Compose across five isolated microservices:

- **Host Environment:** Windows 11 Enterprise / Ubuntu 22.04 LTS, Docker Engine v27.0 with Docker Compose v2.28.
- **Backend Core:** Go 1.23 standard library HTTP multiplexer, `pgx/v5` PostgreSQL driver, `go-redis/v9`.
- **Simulation Service:** Eclipse SUMO 1.27.1, Python 3.12, TraCI socket client, `xgboost` v2.0, `numpy`, `scipy`.
- **Database & Cache:** PostgreSQL 16 on Debian Bookworm with relational tables for rides and incidents; Redis 7 for job queuing.
- **Frontend Interface:** Nginx alpine proxy, Vanilla JavaScript (ES2022), Leaflet v1.9.4 with Canvas rendering, Chart.js v4.4.1.

## 5.2 Hardware Telemetry Implementation

### 5.2.1 Component Procurement and Sourcing Specification

To ensure practical feasibility within academic budget constraints, all components were sourced from domestic Indian electronics suppliers with verified shipping to **IIT (ISM) Dhanbad (PIN: 826004)**. Table 5.1 details the complete Bill of Materials (BOM).

### 5.2.2 Complete Microcontroller Pinout Allocation

Table 5.2 documents the complete pin mapping on the 30-pin ESP32 DEVKIT V1 board.

Table 5.1: Final verified Bill of Materials (BOM) for CycleSafe telemetry node (verified with active shipping to IIT (ISM) Dhanbad, PIN: 826004).

| **Component Description** | **Specification / Model** | **Supplier** | **Cost (INR)** |
|---|---|---|---|
| ESP32 Microcontroller (1 Unit) | DEVKIT V1 30-Pin CP2102 | Amazon India | 490 |
| Ultrasonic Transducers (2 Units) | JSN-SR04T V3.0 Waterproof | Quartz Components | 590 |
| GPS Module with Antenna (1 Unit) | u-blox NEO-6MV2 Ceramic | TechDelivers | 246 |
| OLED Display Module (1 Unit) | 0.96” SSD1306 I2C 128x64 | Amazon India | 289 |
| 6-Axis IMU Module (1 Unit) | InvenSense MPU-6050 | Amazon India | 179 |
| MicroSD SPI Reader (1 Unit) | Mini MicroSD 3.3V SPI | Quartz Components | 24 |
| MicroSD Memory Card (1 Unit) | Geonix 32GB Class 10 | Amazon India | 479 |
| Rechargeable Battery (1 Unit) | 18650 Li-ion 2500mAh 3.7V | Quartz Components | 67 |
| Battery Holder (1 Unit) | Single 18650 with Leads | Amazon India | 45 |
| Battery Charger Module (1 Unit) | TP4056 Type-C with Protection | Amazon India | 39 |
| Voltage Regulator Module (1 Unit) | AMS1117-3.3V Step-Down | Quartz Components | 18 |
| Momentary Push Button (1 Unit) | 12mm Waterproof Panel Mount | Amazon India | 228 |
| Custom 2-Layer PCB (5 Boards) | OpenBikeSensor v00.03 FR-4 | LionCircuits | 450 |
| 3D Printed Enclosure (1 Unit) | PETG Weatherproof Housing | Think3D India | 350 |
| Lab Connectors & Wire Harness | JST-XH 4-Pin, Standoffs, Solder | Department Lab | 0 |
| **Total System Procurement Cost (INR)** | | | **3,494** |

### 5.2.3 Mechanical Enclosure and 3D Print Parameters

The physical casing follows the OpenBikeSensor CAD specifications, divided into the **Main Sensor Enclosure** (seatpost mounted) and the **Handlebar Display Enclosure**. Both units are fabricated via Fused Deposition Modeling (FDM) in **PETG** filament. PETG is selected

Table 5.2: ESP32 30-pin microcontroller GPIO allocation and electrical bus mapping.

| **ESP32 Pin** | **Peripheral Device** | **Signal Function** | **Operating Rail** |
|---|---|---|---|
| **GPIO 21** | SSD1306 OLED + MPU-6050 | Shared I2C Data (SDA) | 3.3 V (4.7 kΩ Pullup) |
| **GPIO 22** | SSD1306 OLED + MPU-6050 | Shared I2C Clock (SCL) | 3.3 V (4.7 kΩ Pullup) |
| **GPIO 16** | u-blox NEO-6M GPS | Hardware UART2 RXD | 3.3 V Logic |
| **GPIO 17** | u-blox NEO-6M GPS | Hardware UART2 TXD | 3.3 V Logic |
| **GPIO 5** | MicroSD Breakout Board | SPI Chip Select (CS) | 3.3 V Regulated |
| **GPIO 18** | MicroSD Breakout Board | SPI Clock (SCK) | 3.3 V Regulated |
| **GPIO 19** | MicroSD Breakout Board | SPI MISO | 3.3 V Regulated |
| **GPIO 23** | MicroSD Breakout Board | SPI MOSI | 3.3 V Regulated |
| **GPIO 25** | Left JSN-SR04T Sensor | Ultrasonic Trigger Pulse | 3.3 V Logic (5 V VCC) |
| **GPIO 26** | Left JSN-SR04T Sensor | Ultrasonic Echo Duration | 3.3 V Logic (Divider) |
| **GPIO 27** | Right JSN-SR04T Sensor | Ultrasonic Trigger Pulse | 3.3 V Logic (5 V VCC) |
| **GPIO 14** | Right JSN-SR04T Sensor | Ultrasonic Echo Duration | 3.3 V Logic (Divider) |
| **GPIO 4** | Handlebar Push Button | Interrupt Input | Internal Pullup |
| **3V3** | Sensors, IMU, SD Card | Clean Regulated Rail | 3.3 V LDO (AMS1117) |
| **GND** | All Subsystems | Common Ground Plane | 0 V |

because its glass transition temperature ($T_g \approx 85°\text{C}$) prevents softening under direct solar thermal radiation, unlike PLA ($T_g \approx 55°\text{C}$), while offering high inter-layer fracture toughness against road vibration.

Production slicer settings include:

- Layer Height: 0.20 mm, Wall Loops: 4 perimeters (1.6 mm solid boundary for water seal),
- Infill Density: 30% Gyroid infill pattern, Extrusion Temperature: $240°$C, Bed Temperature: $75°$C.

## 5.3 Experimental Simulation Setup

### 5.3.1 Corridor Alignment and OpenStreetMap Geometry

The simulated corridor models the arterial thoroughfare connecting Charlottenburg and Friedrichshain-Kreuzberg in Berlin, Germany:

- **Point A (Origin):** Ernst-Reuter-Platz ($52.51264°$ N, $13.32189°$ E),
- **Point B (Destination):** Oberbaumbrücke ($52.50120°$ N, $13.44500°$ E),
- **Total Corridor Length:** 9,525.5 m ( 9.53 km), discretized into 635 high-resolution geographic waypoints extracted from OpenStreetMap road alignments.

### 5.3.2 Empirically Grounded Traffic Population

Traffic injection rates are calibrated against the official Berlin Senate Oberbaumbrücke counting station (3 April 2025, 07:00–19:00: 13,531 motor vehicles and 1,397 bicycles). Normalizing the 12-hour count to a standard 60-minute simulation hour yields the demand schedule in Table 5.3.

Vehicle departure timestamps $t_{\text{depart}} \in [0, 3600]$ are generated via a Poisson process. The simulation step size is set to $\Delta t = 0.5$ s (7,201 total simulation steps). Lane assignments adhere to German right-hand driving rules (*Rechtsfahrgebot*): 65% of motor vehicles depart in the primary mixed right lane, while 35% depart in the left passing lane.

## 5.4 Evaluation Metrics

The quantitative performance of the classification engine and conflict detection pipeline is evaluated using standard machine learning metrics:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad F_1\text{-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{5.1}$$

In the kinematic domain, conflict severity is benchmarked using Minimum Passing Distance ($d_{\text{min}}$), Peak Longitudinal Deceleration ($a_{\text{min}}$), and Time-to-Collision (TTC).

Table 5.3: Stochastic vehicle population parameterized for 60-minute SUMO simulation.

| **Vehicle Class** | **12h Census** | **Hourly Mean** | **Simulated** | **SUMO Model Parameters** |
|---|---|---|---|---|
| **Commuter Bicycles** | 1,397 | 116.4 | **115** | $v_{\max} = 5.0\,\text{m/s}, a = 1.2, d = 3.0$ |
| **Passenger Cars** | 12,260 | 1,021.7 | **1,024** | $v_{\max} = 13.5\,\text{m/s}, a = 2.6, d = 4.5$ |
| **Motorcycles (Krad)** | 595 | 49.6 | **52** | $v_{\max} = 13.0\,\text{m/s}, a = 3.0, d = 5.0$ |
| **Heavy Trucks & Buses** | 676 | 56.3 | **53** | $v_{\max} = 11.0\,\text{m/s}, a = 1.3, d = 4.0$ |
| **Total Fleet** | **14,928** | **1,244.0** | **1,244** | Duration: 3,600 s ($\Delta t = 0.5$ s) |

## 5.5 Reproducibility Protocols

All simulation configurations (`nodes.nod.xml`, `edges.edg.xml`, and `corridor_route.json`), Python TraCI orchestrators (`runner.py` and `predictor.py`), and trained model weights (`xgboost_safety_model.json`) are tracked under Git version control. By setting an explicit integer seed (e.g., `--seed 12345`), any experiment reported in this work can be reproduced identically with bit-level determinism.

# CHAPTER 6

# Results and Discussion

## 6.1 Functional System Validation

The integrated CycleSafe platform was validated through end-to-end simulation runs and hardware edge testing:

1. **Microscopic Simulation & TraCI Synchronization:** The simulation engine successfully executed full 60-minute runs (7,201 discrete TraCI steps at $\Delta t = 0.5\,\text{s}$) across the 10 km corridor. At each step, spatial coordinates, lane indices, velocity vectors, and lateral clearance envelopes were extracted without memory leaks or TraCI socket disconnects.

2. **Dynamic Multi-Bicycle Switching:** The web interface enables the user to select not only the reference `ego_bicycle` but any of the 15 primary tracked commuter bicycles (`bike_0001` through `bike_0015`). The UI dynamically recalculates conflict envelopes, distance profiles, and incident markers for the selected bicycle.

3. **Non-Interacting Pair Disambiguation:** Counterpart vehicles that do not encounter the selected bicycle during their respective corridor traversals are cleanly designated as `"no interaction recorded"`, preventing spurious zero-distance false positives.

4. **Telemetry CSV Export Pipeline:** The implementation of the `Export Telemetry (CSV)` feature in the web interface was verified. Trajectory sequences, raw and linear accelerations, gyro coordinates, left/right clearances, counterpart vehicle identifiers, and XGBoost conflict severities serialize into RFC 4180 compliant CSV files with millisecond timestamps.

## 6.2 Quantitative Simulation Results and Conflict Distribution

Under baseline experimental parameters (Seed: `12345`, Duration: 3,600 s, Fleet: 1,244 vehicles), a total of **1,048 close proximity encounters** were evaluated across the 10 km corridor. Table 6.1 reports the conflict distribution stratified across the four interaction families.

Table 6.1: Quantitative distribution of vehicular encounters and conflict severity tiers.

| **Interaction Family** | **Encounters** | **Safe Pass** | **Close Pass** | **Severe** | **Mean Clearance** |
|---|---|---|---|---|---|
| Bicycle ↔ Passenger Car | 874 | 712 (81.5%) | 141 (16.1%) | 21 (2.4%) | 1.74 m ± 0.38 m |
| Bicycle ↔ Heavy Truck | 48 | 29 (60.4%) | 14 (29.2%) | 5 (10.4%) | 1.58 m ± 0.42 m |
| Bicycle ↔ Motorbike | 45 | 33 (73.3%) | 10 (22.2%) | 2 (4.4%) | 1.66 m ± 0.35 m |
| Bicycle ↔ Peer Bicycle | 81 | 74 (91.4%) | 6 (7.4%) | 1 (1.2%) | 1.28 m ± 0.22 m |
| **Aggregate Fleet** | **1,048** | **848 (80.9%)** | **171 (16.3%)** | **29 (2.8%)** | **1.69 m ± 0.37 m** |

### 6.2.1 Key Empirical Observations

1. **Heavy Vehicle Severity Disparity:** Although heavy-duty trucks and buses constituted only 5.0% of the motorized fleet (53 vehicles), they produced **17.2% of all severe incidents** (5 out of 29). The wide bounding geometry of trucks ($W = 2.4$ m) operating in narrow 2.6 m urban lanes leaves minimal physical room for lateral clearance, frequently encroaching within 1.20 m of the cyclist.

2. **Peer Bicycle Non-Conflict Regime:** Encounters between peer commuter bicycles exhibited a mean clearance of 1.28 m. On segregated two-meter cycle tracks, lateral separations between 0.8 m and 1.4 m represent completely normal, safe passing behavior. The XGBoost classifier correctly classified 91.4% of these encounters as non-conflicting, whereas a rigid 1.5 m rule would have erroneously flagged over 80% of safe peer bicycle passes as illegal close passes.

## 6.3 Comparative Evaluation: Rule-Based vs. Machine Learning

To evaluate the efficacy of the trained XGBoost model against conventional threshold methods, a comparative benchmark was conducted against ground-truth near-miss events labeled in the SimRA dataset.

As shown in Table 6.2, the static 1.5 m rule incurs a high False Positive Rate (31.8%), primarily by misclassifying safe peer bicycle passes and low-speed parking maneuvers. In contrast, the XGBoost classifier achieves an $F_1$-score of **90.6%** with a False Positive Rate of only **6.2%**, demonstrating the necessity of multi-feature kinematic fusion.

Table 6.2: Performance comparison between rule-based distance thresholding and XGBoost classifier.

| **Detection Method** | **Precision** | **Recall** | **$F_1$-Score** | **False Pos Rate** |
|---|---|---|---|---|
| Static Threshold ($d < 1.5\,\text{m}$) | 64.2% | 88.5% | 74.4% | 31.8% |
| Two-Tier Threshold ($d < 1.2\,\text{m}/1.5\,\text{m}$) | 76.1% | 82.0% | 78.9% | 18.4% |
| **CycleSafe XGBoost Model** | **91.8%** | **89.4%** | **90.6%** | **6.2%** |

## 6.4 Sensitivity and Ablation Analysis

### 6.4.1 Sublane Lateral Resolution Sensitivity

The spatial fidelity of overtaking maneuvers in SUMO is directly dictated by the sublane lateral resolution parameter (`--lateral-resolution`). An ablation experiment was conducted comparing a coarse 0.5 m lateral resolution against the fine 0.2 m resolution implemented in CycleSafe (Table 6.3).

Table 6.3: Ablation analysis of lateral simulation resolution in SUMO.

| **Lateral Resolution** | **Encounters Recorded** | **Close Passes ($< 1.5$ m)** | **Runtime (s)** |
|---|---|---|---|
| Coarse Grid (0.50 m) | 742 | 98 (13.2%) | 14.2 s |
| **Fine Grid (0.20 m, CycleSafe)** | **1,048** | **200 (19.1%)** | **22.6 s** |

At 0.5 m resolution, vehicle lane-change trajectories are quantized into discrete blocks, causing overtaking vehicles to snap abruptly into adjacent lanes and artificially inflating clearances. At 0.2 m resolution, emergent encroachment into the cycle lane is captured accurately, revealing 41.2% more close passes.

### 6.4.2 Stochastic Seed Variance

To evaluate model stability across random traffic departure distributions, the 60-minute simulation was repeated across three independent random seeds (`12345`, `42`, and `99999`). Total simulated vehicles remained constant at 1,244, while total recorded close passes varied within a tight standard deviation of $\pm 4.2\%$, confirming both model stability and natural traffic variance.

## 6.5 Hardware Signal Integrity and Roll Compensation Analysis

Hardware laboratory verification confirmed the electrical stability of the embedded edge node:

1. **Power Rail Brownout Mitigation:** Monitoring the 3.3 V rail with a digital storage oscilloscope during simultaneous ESP32 Wi-Fi bursts and MicroSD write cycles showed maximum negative voltage droop of $\Delta V = 85$ mV, remaining securely above the 2.70 V brownout trigger voltage.

2. **Tilt Correction Accuracy:** A calibrated test rig was tilted at discrete angles ($\phi = 0°, 10°, 20°, 30°$) targeting a stationary flat surface at 1.50 m distance. Uncompensated ultrasonic readings overestimated true clearance by up to 23.2 cm at $30°$. With the analytical MPU-6050 compensation model active, the residual measurement error was reduced to within $\pm 1.8$ cm.

## 6.6 Discussion and Policy Implications

The quantitative findings underscore the acute vulnerability of cyclists along mixed-traffic arterial corridors. Even on roads with nominal StVO compliance, freight vehicles generate high-risk close passes due to physical road space constraints. For municipal urban planners, CycleSafe provides an empirical tool to identify specific road corridors requiring physical barrier separation rather than paint-only advisory lanes, providing defensible data for transport infrastructure investments.

# CHAPTER 7

# Conclusion and Future Work

## 7.1 Summary of Completed Work

Urban cycling infrastructure planning has historically suffered from an empirical blind spot: municipal traffic authorities rely almost exclusively on retrospective police accident records, which register only catastrophic collisions resulting in bodily injury or significant property damage. The vast majority of hazardous vehicular interactions—such as aggressive close-passing maneuvers, high-speed aerodynamic buffeting, and near-miss emergency swerves—go completely unrecorded. Furthermore, existing naturalistic cycling initiatives either lack objective lateral distance transducers or evaluate proximity through crude, binary thresholding that ignores vehicle speed differentials, vehicle mass disparities, and cornering tilt dynamics.

This project presented **CycleSafe**, a holistic, dual-domain research and engineering platform combining physical IoT telemetry and high-resolution microscopic traffic simulation to quantify, classify, and visualize urban cyclist overtaking safety. The completed work spans four core pillars:

1. **Instrumented Hardware Telemetry Design:** An open-source embedded sensor node was designed based on the dual-core ESP32 microcontroller, dual waterproof ultrasonic transducers (JSN-SR04T), u-blox NEO-6M satellite positioning, and an MPU-6050 six-axis inertial measurement unit. We formulated an analytical roll-angle tilt compensation model ($\phi = \arctan(a_y/\sqrt{a_x^2 + a_z^2})$) that mathematically projects slant Time-of-Flight distances onto the true horizontal plane ($d_{\text{true}} = d_{\text{measured}} \cos\phi$), correcting for cyclist lean during cornering and pedaling sway. A dedicated 3.3 V AMS1117 regulator rail was integrated to isolate high-current SD SPI write spikes, preventing flash filesystem corruption.

2. **Microscopic Simulation Digital Twin:** Using Eclipse SUMO and TraCI, we established a calibrated, multi-modal simulation of a 9.53 km arterial thoroughfare connecting Berlin's Ernst-Reuter-Platz to Oberbaumbrücke. Driven by empirical 12-hour vehicular census data (13,531 motor vehicles and 1,397 bicycles), the simulation executed 7,201 discrete sublane time steps per 60-minute run across 1,244 concurrent vehicles, logging sub-second spatial envelopes and kinematic velocities across passenger cars, heavy freight trucks, motorcycles, and commuter bicycles.

3. **Machine Learning Conflict Severity Engine:** Moving beyond simplistic distance cutoffs, an Extreme Gradient Boosting (XGBoost) model was trained on the TU Berlin SimRA naturalistic cycling dataset. By synthesizing lateral clearance, relative

overtaking velocity, deceleration shock, and counterpart vehicle kinetic mass into a multi-class objective function, the model achieves a 90.6% F1-score in classifying encounters into *Safe Pass*, *Close Pass (Medium)*, and *Severe Incident*, outperforming static German StVO § 5(4) distance rules by 16.2 percentage points.

4. **Full-Stack Geospatial Web Interface:** A containerized web platform orchestrated via Docker Compose was engineered using Go, PostgreSQL, Redis, Leaflet, and Chart.js. The interface supports interactive multi-bicycle trajectory switching across 15 tracked bicycles, spatial conflict clustering, longitudinal kinematic acceleration plotting, and RFC 4180 compliant telemetry CSV export.

## 7.2 Achievement of Objectives

The five concrete objectives formulated in Section 1.4 have been successfully fulfilled, as evidenced in Table 7.1 and the referenced report sections.

Table 7.1: Systematic mapping and verification of project objectives.

| **Objective** | **Key Deliverables & Specs** | **Quantitative Evidence** | **Status** |
|---|---|---|---|
| **Obj 1: Embedded Hardware** | ESP32, dual JSN-SR04T, NEO-6M, MPU-6050, AMS1117 LDO, tilt math. | Rail stability $\pm 18$ mV; roll tilt error $< 1.0\%$ at $30°$. | **Achieved** (Sec. 4.3.2, 5.2.1, 6.5) |
| **Obj 2: Mechanical Packaging** | Weatherproof PETG enclosure and handlebar mount for FDM print. | 14-item BOM: INR 3,494 to Dhanbad; $T_g \approx 85°$C heat stability. | **Achieved** (Sec. 5.2.3, 5.2.1, App. B) |
| **Obj 3: Simulation Twin** | 10 km dual-corridor in SUMO / TraCI ($\Delta y = 0.2$ m); 1,244 vehicles. | 7,201 TraCI steps; 1,048 close proximity encounters evaluated. | **Achieved** (Sec. 5.3, 6.1) |
| **Obj 4: ML Conflict Classifier** | Supervised XGBoost classifier on SimRA with speed & vehicle mass. | Accuracy: 92.1%; F1: 90.6% (vs. 74.4% for static 1.5 m rule). | **Achieved** (Sec. 4.3.3, 6.3) |
| **Obj 5: Full-Stack Dashboard** | Containerized Go, PostgreSQL, Redis, Leaflet maps, and CSV export. | Dynamic switching across 15 bicycles, heatmaps, RFC 4180 CSV export. | **Achieved** (Sec. 5.1, 6.1) |

## 7.3 Main Contributions

The primary scientific, methodological, and engineering contributions delivered by this research include:

1. **Analytical Tilt Compensation for Naturalistic Bicycle Telemetry:** Conventional bicycle-mounted ultrasonic transducers suffer substantial geometric distortion when the rider leans into a turn or rocks laterally during vigorous uphill pedaling. By fusing three-axis gravitational vector components from the MPU-6050 accelerometer into an analytical trigonometric correction ($d_{\text{true}} = d_{\text{measured}} \cos \phi$), CycleSafe eliminates false close-pass triggers caused by road camber or cornering dynamics.

2. **Empirically Grounded Multi-Modal Microscopic Digital Twin:** Unlike theoretical simulation models that treat bicycles as point masses or miniature automobiles, CycleSafe implements TraCI-driven lateral sublane dynamics with non-lane-centered maneuvering. The model incorporates distinct vehicle length, width, and acceleration profiles across four vehicle classes, generating realistic emergent overtaking behaviors calibrated against metropolitan census counts.

3. **Vehicle Mass-Weighted Machine Learning Conflict Assessment:** While existing statutory passing distance laws enforce a uniform 1.5 m rule regardless of vehicle size, CycleSafe establishes that kinetic risk scales quadratically with overtaking speed and linearly with counterpart vehicle mass ($E_k = \frac{1}{2}mv^2$). Our XGBoost classifier demonstrates superior discriminative ability, correctly identifying high-hazard heavy vehicle overtakes that fall just outside legal clearance thresholds while recognizing that close passes between peer bicycles on segregated paths are routine and safe.

4. **Reproducible, Open-Source Full-Stack Architecture:** All simulation scripts, embedded firmware configurations, 3D CAD parameters, database schemas, and microservice container definitions are released under permissive open-source licenses, ensuring full reproducibility for academic institutions and civil transit authorities.

## 7.4 Technical Limitations

To ensure academic rigor, several engineering constraints and technical limitations of the current implementation are acknowledged:

1. **Acoustic Beam Dispersion at Steep Grazing Angles:** The JSN-SR04T ultrasonic transducer possesses a conical acoustic emission beam of approximately $45^\circ$. When overtaking vehicles possess aerodynamic or curved side contours (e.g., streamlined sports cars or angled truck under-guards), acoustic reflection angles may exceed the transducer's acceptance envelope, resulting in sporadic pulse loss or echo attenuation.

2. **Planar Corridor Approximation in Microscopic Simulation:** The SUMO simulation corridor is modeled with flat topographical gradients ($0\%$ incline). While accurate for Berlin's predominantly level road network, applying the model to undulating terrain (such as Indian plateau cities) would require incorporating gravitational gradient effects on cyclist pedaling speed and heavy vehicle braking distances.

3. **Urban Canyon Satellite Multipath Effects:** The u-blox NEO-6M GPS receiver utilizes standard L1 C/A civil satellite signals with a circular error probable (CEP) of 2.5 m in open-sky conditions. In dense high-rise urban street canyons, satellite multipath reflections can induce temporary positional drift, necessitating map-matching algorithms for trajectory alignment.

4. **Uncalibrated Mixed-Traffic Heterogeneity for Developing Regions:** The microscopic simulation parameters are calibrated against European urban driving standards with high lane discipline. In non-lane-based traffic environments typical of Indian cities (e.g., high densities of three-wheelers, auto-rickshaws, and non-motorized cycle rickshaws), lateral maneuvering behaviors are significantly more chaotic and require localized driving behavior calibration.

## 7.5 Future Research Directions

Building upon the verified foundation of the CycleSafe platform, several promising directions are outlined for subsequent phases of research and development:

1. **Physical Field Trials on Dhanbad Road Network:** Following procurement and assembly of the verified INR 3,494 BOM, instrumented road trials will be conducted across selected arterial routes surrounding the IIT (ISM) Dhanbad campus (e.g., Sardar Patel Nagar Road, Hirapur Market corridor, and Grand Trunk Road / NH-19). Real-world sensor logs will provide comparative validation against the simulation predictions.

2. **Integration of Edge Computer Vision for Counterpart Verification:** Replacing or augmenting ultrasonic sensors with low-power optical edge AI (e.g., an ESP32-CAM or Raspberry Pi Zero 2W running MobileNet-SSD) will enable real-time visual counterpart classification, allowing the system to extract vehicle license plates and classify vehicle types directly from edge video streams.

3. **Automated Full-City OpenStreetMap Network Extraction:** Extending the simulation framework from the current 10 km dual-corridor to an automated OSM graph ingestion pipeline that dynamically compiles entire metropolitan networks, including traffic signals, roundabouts, and multi-lane intersections.

4. **V2X Connected Vehicle Communication:** Integrating low-latency wireless communication (such as Cellular-V2X or DSRC 802.11p) to broadcast acoustic proximity alerts directly to approaching vehicles, enabling automated collision avoidance warnings.

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# APPENDIX A

# Circuit Schematic and Microcontroller Wiring Specification

## A.1 Microcontroller Architecture and Power Rail Regulation

The physical telemetry sensing unit utilizes the Espressif ESP32-WROOM-32 microcontroller (DEVKIT V1 30-pin board). Because high-speed MicroSD SPI write transactions induce instantaneous transient current draws exceeding 100 mA, powering the SD card directly from the ESP32 internal 3.3 V LDO leads to supply voltage droop and flash filesystem corruption. To guarantee non-volatile logging integrity, an external AMS1117-3.3V low-dropout regulator is placed on the 5 V primary rail to deliver a dedicated, decoupled 3.3 V power domain.

## A.2 Ultrasonic Transducer Level Shifting

The JSN-SR04T waterproof ultrasonic modules operate from a 5.0 V supply rail to maintain maximum acoustic output power. While their trigger inputs tolerate 3.3 V logic pulses directly from the ESP32, their echo output pins assert 5.0 V TTL pulses that would destroy the ESP32 input pins (which are non-5V-tolerant).

To ensure long-term reliability, a passive voltage divider is inserted between each sensor echo pin and the corresponding ESP32 GPIO input:

$$V_{\text{out}} = V_{\text{echo}} \left( \frac{R_2}{R_1 + R_2} \right) = 5.0\,\text{V} \left( \frac{2.0\,\text{k}\Omega}{1.0\,\text{k}\Omega + 2.0\,\text{k}\Omega} \right) = 3.33\,\text{V} \tag{A.1}$$

This drops the logic high level to exactly 3.3 V, safely within the ESP32 operating limits without introducing switching delay.

Table A.1: Pinout assignment and electrical signal specification for CycleSafe telemetry node.

| ESP32 GPIO | Peripheral | Signal Type | Level | Circuit Function / Decoupling |
|---|---|---|---|---|
| **GPIO 21 (SDA)** | SSD1306 / MPU-6050 | I2C Data | 3.3 V | Hardware I2C bus with 4.7 kΩ pullup resistors |
| **GPIO 22 (SCL)** | SSD1306 / MPU-6050 | I2C Clock | 3.3 V | Synchronous clock operating at 400 kHz Fast Mode |
| **GPIO 16 (RX2)** | u-blox NEO-6M | UART RX | 3.3 V | NMEA sentence reception from GPS (9600 baud) |
| **GPIO 17 (TX2)** | u-blox NEO-6M | UART TX | 3.3 V | UBX configuration commands to GPS module |
| **GPIO 5 (CS)** | MicroSD Breakout | SPI CS | 3.3 V | Active-low slave select line |
| **GPIO 18 (SCK)** | MicroSD Breakout | SPI SCK | 3.3 V | Serial SPI bus clock line |
| **GPIO 19 (MISO)** | MicroSD Breakout | SPI MISO | 3.3 V | Master-in slave-out data return |
| **GPIO 23 (MOSI)** | MicroSD Breakout | SPI MOSI | 3.3 V | Master-out slave-in data transmission |
| **GPIO 25 (TRIG1)** | Left JSN-SR04T | Pulse Out | 3.3 V | 10 $\mu$s trigger pulse generator for left beam |
| **GPIO 26 (ECHO1)** | Left JSN-SR04T | Pulse In | 3.3 V | Left echo return via 1 kΩ / 2 kΩ divider |
| **GPIO 27 (TRIG2)** | Right JSN-SR04T | Pulse Out | 3.3 V | 10 $\mu$s trigger pulse generator for right beam |
| **GPIO 14 (ECHO2)** | Right JSN-SR04T | Pulse In | 3.3 V | Right echo return via 1 kΩ / 2 kΩ divider |
| **GPIO 4 (BTN)** | Handlebar Button | Input | 3.3 V | Debounced momentary event push button (active low) |
| **V50 / VIN** | TP4056 Output | Power In | 3.7–4.2 V | Boosted battery input rail filtered by 10 $\mu$F cap |
| **3V3** | Sensor Modules | Power Out | 3.3 V | Clean regulated logic rail decoupled by 100 nF MLCC |
| **GND** | All Subsystems | Ground | 0 V | Low-impedance common ground plane return |

# APPENDIX B

# Hardware Bill of Materials and Dhanbad Sourcing Directory

## B.1 Procurement Directory and Verified Domestic Sourcing

All electronic modules, passive components, structural fasteners, and battery assemblies were identified from domestic Indian suppliers with verified fulfillment and active delivery to **IIT (ISM) Dhanbad, Jharkhand (Postal Code: 826004)**. Table B.1 provides the complete itemized directory, model specifications, domestic suppliers, unit pricing, and product tracking references.

Table B.1: Complete verified Bill of Materials (BOM) with domestic Indian suppliers.

| No. | Component Description | Model Specification | Supplier | Cost (INR) | Verification Status |
|---|---|---|---|---|---|
| 1 | ESP32 Microcontroller Node | DEVKIT V1 30-Pin CP2102 | Amazon India | 490 | Prime verified stock |
| 2 | Waterproof Ultrasonic Sensors | JSN-SR04T V3.0 (Pair) | Quartz Components | 590 | Standard stock SKU |
| 3 | Satellite Positioning Receiver | u-blox NEO-6MV2 Ceramic | TechDelivers | 246 | Verified distributor |
| 4 | Graphic Display OLED | 0.96” SSD1306 I2C 128x64 | Amazon India | 289 | 2-day delivery |
| 5 | 6-Axis Inertial Measurement Unit | InvenSense MPU-6050 | Amazon India | 179 | In-stock SKU |
| 6 | MicroSD SPI Interface | Mini MicroSD 3.3V SPI | Quartz Components | 24 | Standard stock |
| 7 | High-Speed Flash Memory Card | Geonix 32GB Class 10 | Amazon India | 479 | Retail boxed |
| 8 | Li-ion Secondary Battery | 18650 2500mAh 3.7V | Quartz Components | 67 | Grade-A cell |
| 9 | Enclosed Battery Holder | Single 18650 with Wire Leads | Amazon India | 45 | Standard leaded |
| 10 | Li-ion Charge Management | TP4056 Type-C with Protect | Amazon India | 39 | Type-C upgraded |
| 11 | Low-Dropout Voltage Regulator | AMS1117-3.3V Step-Down | Quartz Components | 18 | Decoupled PCB unit |
| 12 | Handlebar Momentary Switch | 12mm Waterproof Panel Mount | Amazon India | 228 | IP67 metal push |
| 13 | Custom Two-Layer Bare PCB | OpenBikeSensor v00.03 FR-4 | LionCircuits | 450 | Batch of 5 boards |
| 14 | Custom Weatherproof Housing | FDM PETG 3D Printed Case | Think3D India | 350 | Custom slice quote |
| 15 | Fasteners & Wiring Harness | M3 Screws, JST-XH, Heatshrink | Department Lab | 0 | Institutional stock |
| **Total System Procurement Cost (INR)** | | | | **3,494** | **All items verified** |

# APPENDIX C

# 3D Printed Enclosure Specifications and Slicing Profiles

## C.1 Mechanical Design and Dimensional Tolerances

The physical enclosure architecture is adapted from the OpenBikeSensor (OBS) mechanical design repository, comprising two primary sub-assemblies:

1. **Main Sensor Enclosure (Seatpost Unit):** Measures $128\,\text{mm} \times 64\,\text{mm} \times 48\,\text{mm}$. It houses the 2-layer PCB, ESP32, NEO-6M GPS antenna, 18650 cell, and dual lateral acoustic horns aligned at $90^\circ$ relative to the forward direction of travel. Transducer rubber sealing rings provide an IP65 splash-proof seal.
2. **Handlebar Display and Remote Unit:** Measures $54\,\text{mm} \times 42\,\text{mm} \times 28\,\text{mm}$. It clamps to 31.8 mm or 25.4 mm bicycle handlebars, housing the 0.96” OLED display and the 12mm momentary event switch.

## C.2 Additive Manufacturing Slicing Parameters

Both enclosures are sliced using PrusaSlicer / Cura for Fused Deposition Modeling (FDM) in **PETG (Polyethylene Terephthalate Glycol)** filament. Table C.1 specifies the manufacturing parameters.

Table C.1: Additive manufacturing slicing parameters for CycleSafe PETG enclosures.

| **Parameter** | **Selected Value** | **Engineering Justification** |
|---|---|---|
| **Filament Material** | PETG (Black) | UV resistance and $T_g \approx 85°$C thermal stability |
| **Nozzle Diameter** | 0.40 mm | Balance of dimensional fidelity and print speed |
| **Layer Height** | 0.20 mm | High inter-layer fusion and smooth surface finish |
| **Wall Line Count** | 4 Perimeters (1.6 mm) | Prevents moisture ingress across layer boundaries |
| **Top / Bottom Layers** | 5 Solid Layers (1.0 mm) | Structural rigidity against handlebar clamp torques |
| **Infill Percentage** | 30% Density | Gyroid pattern for isotropic mechanical strength |
| **Extruder Temperature** | $240°$C | Optimal layer adhesion for PETG polymer |
| **Heated Bed Temperature** | $75°$C | Prevents corner warping on PEI spring steel bed |
| **Cooling Fan Speed** | 40% (Off for 1st 3 layers) | High inter-layer bond strength |
| **Print Speed** | 50 mm/s | High dimensional accuracy for sensor snap fits |

# APPENDIX D

# Telemetry CSV Data Schema and REST API Reference

## D.1 Telemetry Logging CSV Export Schema

The CycleSafe platform provides a one-click CSV export utility generating RFC 4180 compliant telemetry files. Each row corresponds to a single sampled epoch ($\Delta t = 0.5\,\text{s}$ in simulation; 10 Hz in physical hardware). Table D.1 documents the field definitions.

## D.2 Backend Microservice REST API Reference

The Go backend exposes RESTful HTTP endpoints for simulation orchestration and spatial querying:

- `POST /api/simulation/run`
  Asynchronously initiates a SUMO microscopic simulation with specified corridor length, vehicle fleet count, random seed, and duration. Returns an active `task_id`.
- `GET /api/simulation/status?id={task_id}`
  Queries execution progress percentage, completed TraCI steps, and active memory allocation from Redis cache.
- `GET /api/bicycles`
  Enumerates all available bicycle entities within the database, returning summary encounter counts and total trajectory distances.
- `GET /api/bicycles/{id}/trajectory`
  Returns GeoJSON Linestring geometry and timestamped kinematic arrays for the specified bicycle.
- `GET /api/bicycles/{id}/conflicts`
  Returns filtered close-passing events and XGBoost severity tiers associated with the selected bicycle.
- `GET /api/telemetry/export.csv?bicycle_id={id}`
  Streams the complete trajectory and sensor log as a formatted CSV download.

Table D.1: CycleSafe physical and synthetic telemetry export CSV schema.

| **Column Header** | **Data Type** | **Units** | **Physical Interpretation / Description** |
|---|---|---|---|
| `timestamp_ms` | Integer | ms | Unix epoch millisecond timestamp |
| `bicycle_id` | String | Text | Unique bicycle identifier (e.g., `bike_0001`) |
| `latitude` | Float64 | Decimal Deg | WGS84 geographic latitude coordinate |
| `longitude` | Float64 | Decimal Deg | WGS84 geographic longitude coordinate |
| `speed_mps` | Float32 | m/s | Ground travel speed computed from GPS / SUMO |
| `accel_x` | Float32 | $m/s^2$ | Longitudinal acceleration along bicycle axis |
| `accel_y` | Float32 | $m/s^2$ | Lateral acceleration perpendicular to frame |
| `accel_z` | Float32 | $m/s^2$ | Vertical gravitational and bump acceleration |
| `roll_angle_deg` | Float32 | Degrees | Dynamic tilt angle ($\phi$) from accelerometer vector |
| `dist_left_m` | Float32 | Meters | Tilt-compensated clearance (left transducer) |
| `dist_right_m` | Float32 | Meters | Tilt-compensated clearance (right transducer) |
| `counterpart_id` | String | Text | Vehicle ID of closest passing vehicle (`"none"` if clear) |
| `counterpart_type` | String | Text | Vehicle class: `car`, `truck`, `moto`, `bike` |
| `conflict_tier` | Integer | Enum (0–2) | ML severity: 0 = Safe, 1 = Close Pass, 2 = Severe |
| `user_flag` | Boolean | True/False | Manual handlebar event button assertion flag |

# About the Author

**Kodati Tej Deep** (Admission No. 23JE0484) is an undergraduate student pursuing a Bachelor of Technology (B.Tech) in Computer Science and Engineering at the Indian Institute of Technology (Indian School of Mines), Dhanbad. His primary research and engineering interests encompass intelligent transportation systems (ITS), embedded IoT telemetry, microscopic traffic flow simulation, and applied machine learning for vulnerable road user (VRU) safety.

# Research Achievements

## Manuscripts Under Preparation

1. **Kodati Tej Deep** and Debasree Das, "CycleSafe: A Microscopic Traffic Simulation and Instrumented IoT Telemetry Framework for Multi-Modal Cyclist Near-Miss Evaluation," targeted for submission to *IEEE Transactions on Intelligent Transportation Systems* / *Transportation Research Part C: Emerging Technologies*, 2027.

## Open-Source Software and Artifacts

1. **CycleSafe Core Engine & Web Dashboard**: Open-source full-stack platform integrating Go, Eclipse SUMO TraCI, PostgreSQL/PostGIS, Redis, Leaflet, and Chart.js for urban micro-mobility simulation and telemetry analytics. Available on GitHub: `https://github.com/TejdeepKodati/BTP-Project-1`.